

Generation Interconnection System Impact Study Report

For Queue Project Q14-0379

Albreda Tans Mtn Tap 138 kV

October 29, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0379. The Interconnection Customer proposes a 74.7 MW solar photovoltaic facility interconnecting at the Albreda Tans Mtn Tap 138 kV substation in the Virginia area. The request was received on September 16, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 160 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$359,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Remington Ct to Riverton 230 Substation 138.0 kV
Contingency: Tuckasegee Tie Substation to Merced Substation 138.0 kV Line
Loading: 106.5% (Rating: 499 MVA)
DFAX: 6.2%
Pre-project loading: 105.6%
2. Facility: Birchwood to Nauvoo 345.0 kV Line
Contingency: Dunn Siding to Fredericksburg 345.0 kV
Loading: 100.1% (Rating: 1,663 MVA)
DFAX: 8.7%
Pre-project loading: 99.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Remington Ct to Riverton 230 Substation 138.0 kV	Reconductor line	\$8,390,000	2% (\$129,000)
Nauvoo to Birchwood 345.0 kV	Reconductor line	\$27,600,000	1% (\$230,000)
	Total allocated to project		\$359,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.